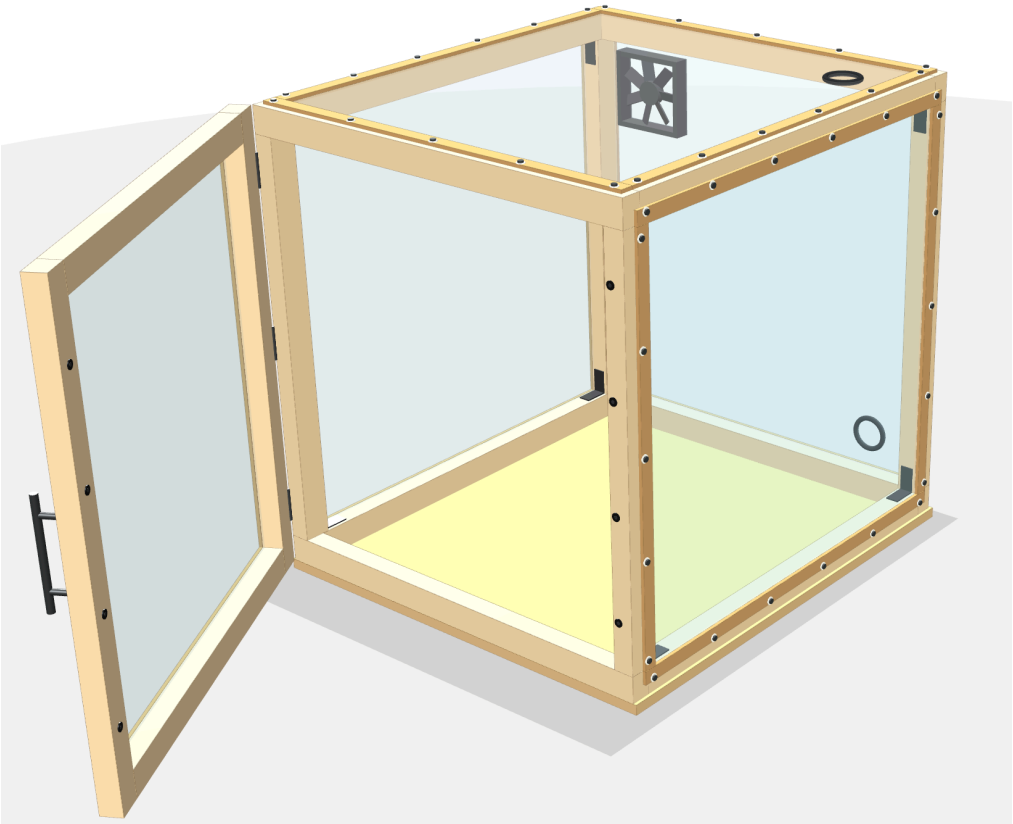
# Bambu Lab A1 Wooden Enclosure

Timber machining drawings • external feed (AMS lite outside) + rebate-and-bead version



3D model view (door half open, printer hidden)

| | |
|---|---|
| Frame (outside of wood) | **640 × 700 × 680** |
| Clear inside W×D×H | **580 × 640 × 640** |
| Finished size (door, beads, feet) | **about 650 × 743 × 710** |
| 30×40 strip total | **10.0 m (6 × 2 m)** |
| 20×5 bead total | **12.4 m (7 × 2 m)** |

## General notes

1. All dimensions in millimetres (mm).
2. Timber: 30×40 kiln-dried spruce or pine, planed four sides. Cut the rebates yourself with a router or use a joinery shop that can do stopped rebates; most timber yards only cut to length, in which case use the plain-strip drawings.
3. Tolerances: length ±1, section ±0.5; rebate width 10 (+0.5 / 0), depth 4 (±0.3).
4. Every rebate is 10 wide × 4 deep, so a 3 mm panel on 1 mm foam sits flush with the wood.
5. Profiles (sheet 3): P0 no rebate; PA one rebate in a 40 face; PB one rebate in a 30 face; PC two rebates, one in a 40 face and one in a 30 face, on diagonally opposite corners.
6. "Full length" = the rebate runs end to end. "Stopped" = solid wood is left at both ends; square the ends with a chisel.
7. Cut to length first, then rebate. Pencil the part number (A1, B2 …) on every piece.
8. Orange hatching = rebate; blue dashed outline = clear panel.

## Sheet list


1. Overview
2. Cut list
3. Profiles and panel detail
4. Frame parts
5. Door frame
6. Frame assembly views
7. Clear panels
8. Appendix: screw-on version


Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

**Overview**

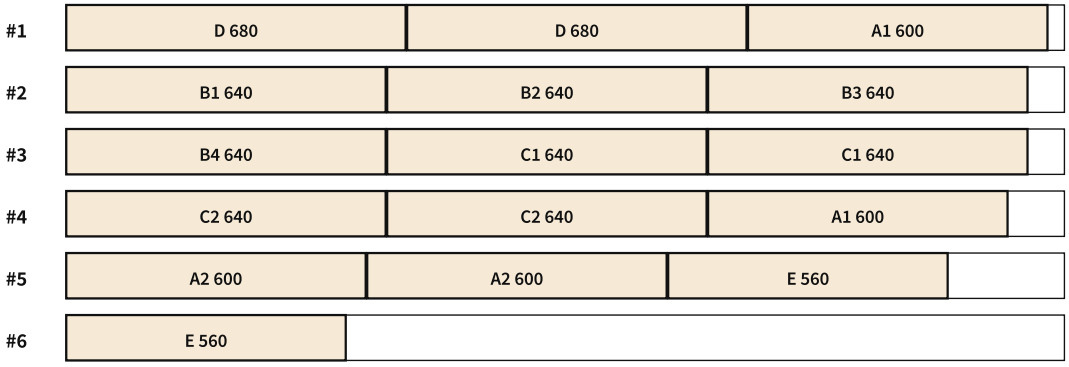
| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 1 / 8 | — | mm | 2026-11-08 | v2 |

## Cut list • 30×40 strip

| ID | Part | Size (length) | Profile | Rebate | Qty | Location |
|---|---|---|---|---|---|---|
| **A1** | Front post | 30×40×600 | PA | Full length | 2 | Front left & right |
| **A2** | Rear post | 30×40×600 | PC | Full length (×2) | 2 | Rear left & right |
| **B1** | Bottom front rail | 30×40×640 | P0 | — | 1 | Bottom frame, front |
| **B2** | Bottom back rail | 30×40×640 | PA | Stopped 20–620 | 1 | Bottom frame, back |
| **B3** | Top front rail | 30×40×640 | PB | Stopped 20–620 | 1 | Top frame, front |
| **B4** | Top back rail | 30×40×640 | PC | Stopped 20–620 (×2) | 1 | Top frame, back |
| **C1** | Bottom side rail | 30×40×640 | PA | Full length | 2 | Bottom frame, left & right |
| **C2** | Top side rail | 30×40×640 | PC | Full length (×2) | 2 | Top frame, left & right |
| **D** | Door stile | 30×40×680 | PA | Stopped 30–650 | 2 | Door left & right |
| **E** | Door rail | 30×40×560 | PA | Full length | 2 | Door top & bottom |
| **F** | Base board | 650×705×15 | — | — | 1 | Paulownia panel, cut to size only |

**Strip total 10 000 mm, 16 pieces (A1–E), plus 1 base board.**

## Cutting plan • 2 m stock, 3 mm kerf

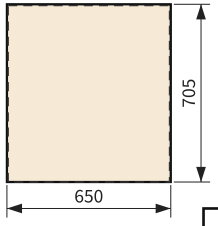



## G Glazing bead • solid wood 20×5

| Length | Qty | Use |
|---|---|---|
| **660** | 4 | Side panels, top & bottom (2 panels) |
| **640** | 2 | Top panel, left & right |
| **620** | 4 | Back panel top/bottom 2; top panel front/back 2 |
| **600** | 10 | Side panels front/back 4; back panel sides 2; door all 4 |

Total 12.4 m, buy 7 × 2 m. Paulownia works; ready-made pine or beech strip splits less.
Horizontal beads run across the corners; upright beads fit between them.

## F Base board 1:30




15 mm paulownia panel. Dashed = frame footprint (5 in at left, right and back; flush at the front).

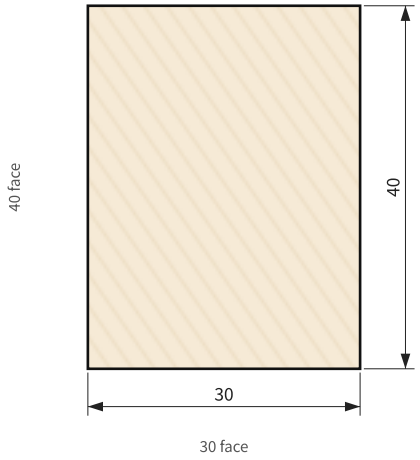
Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

**Cut list**

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 2 / 8 | — | mm | 2026-11-08 | v2 |

## Profiles (scale 1.2:1)

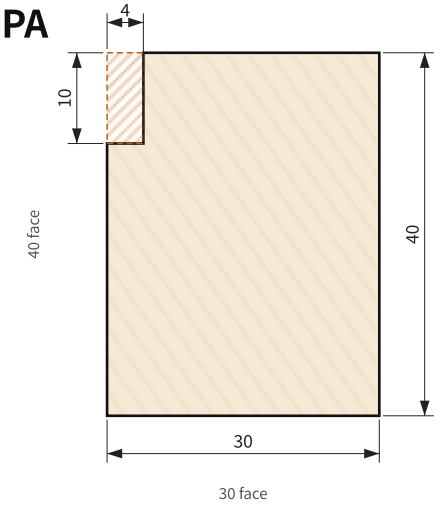
### P0




No rebate

**Used for: B1**

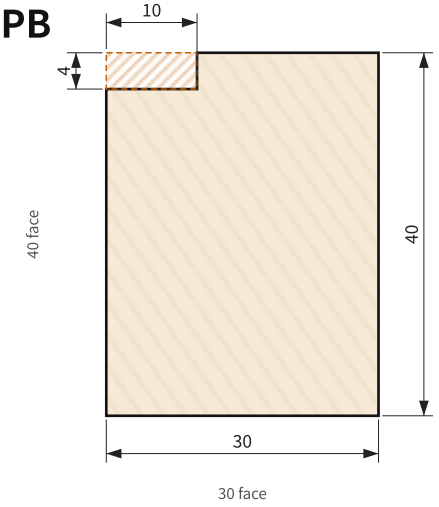
### PA




One rebate in a 40 face: 10 wide, 4 deep

**Used for: A1 B2 C1 D E**

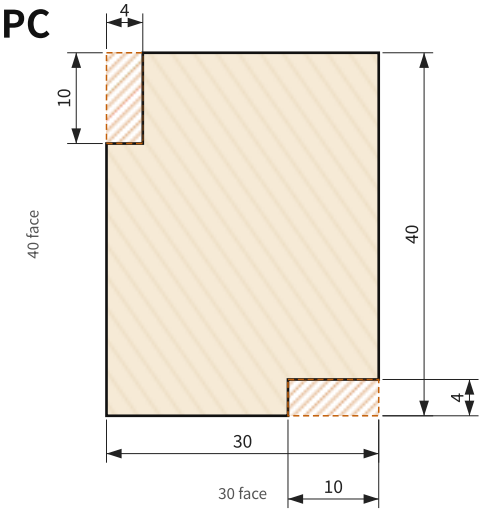
### PB




One rebate in a 30 face: 10 wide, 4 deep

**Used for: B3**

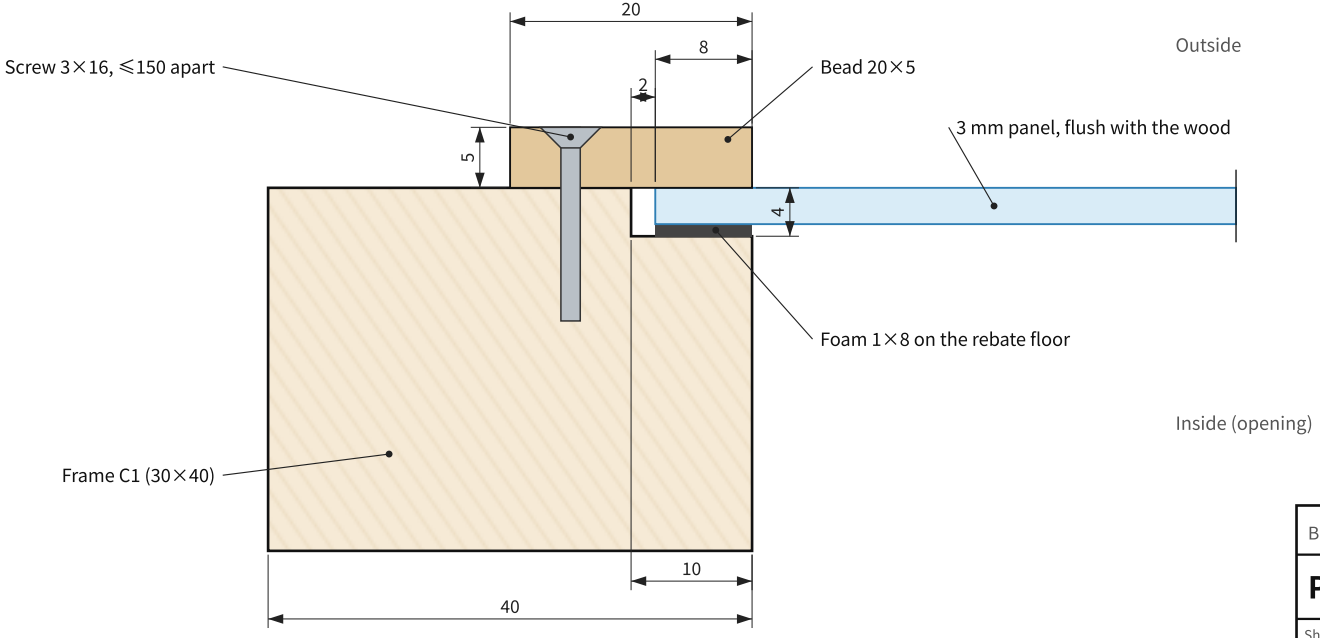
### PC




Two: one in a 40 face + one in a 30 face, diagonal corners

**Used for: A2 B4 C2**

## Panel detail (scale 1.6:1, bottom side rail C1 shown; every rebate works the same way)




2 mm = expansion gap: the panel grows when warm, so it must not be wedged in.

8 mm = how far the panel edge sits in the rebate.

Centre the panel with 2 mm spacers, then remove them once the beads are on.

Bambu Lab A1 enclosure · external feed · rebate + bead (router needed)

# Profiles and panel detail

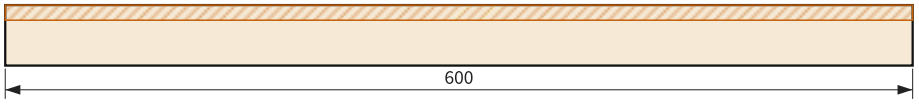
| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 3 / 8 | 1.2:1 / 1.6:1 | mm | 2026-11-08 | v2 |

## Frame parts (A1–C2)

Scale 1:5. Bar = the rebated face; dashed = the second rebate on the opposite corner (far side); small drawing = section.

### A1 Front post

×2 • PA

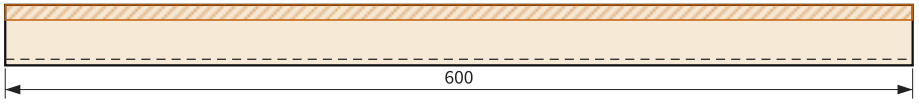


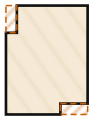



40 face outward (side), rebate on the rear edge. Left and right are the same part.

### A2 Rear post

×2 • PC

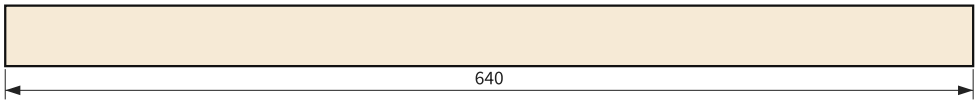


40 face outward (side), rebate on the front edge; 30 face to the back, rebate on the inner edge. Left/right identical, turned end for end.

### B1 Bottom front rail

×1 • P0

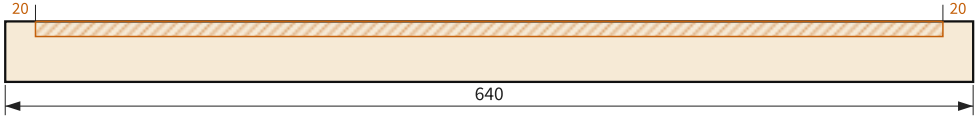





No rebate.

### B2 Bottom back rail

×1 • PA

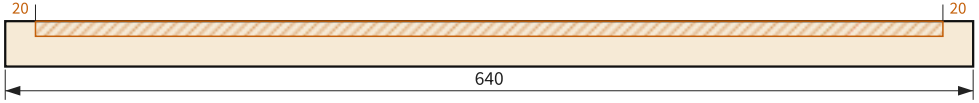





40 face to the back, rebate on the top edge; stopped, 20 solid at each end.

### B3 Top front rail

×1 • PB

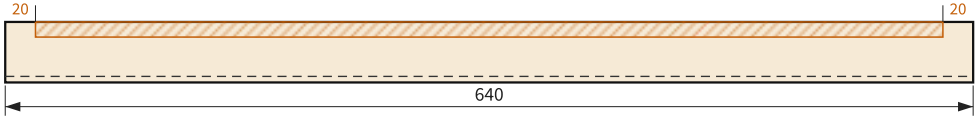


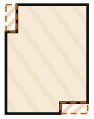



30 face up, rebate on the inner (rear) edge; stopped, 20 solid at each end.

### B4 Top back rail

×1 • PC

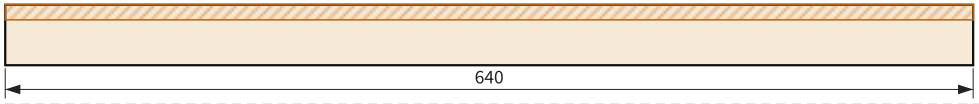


40 face to the back, rebate on the bottom edge; 30 face up, rebate on the inner (front) edge. Both stopped, 20 solid at each end.

### C1 Bottom side rail

×2 • PA

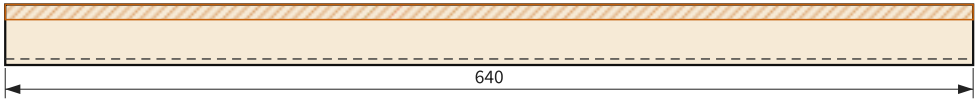


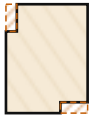



40 face outward, rebate on the top edge; full length.

### C2 Top side rail

×2 • PC



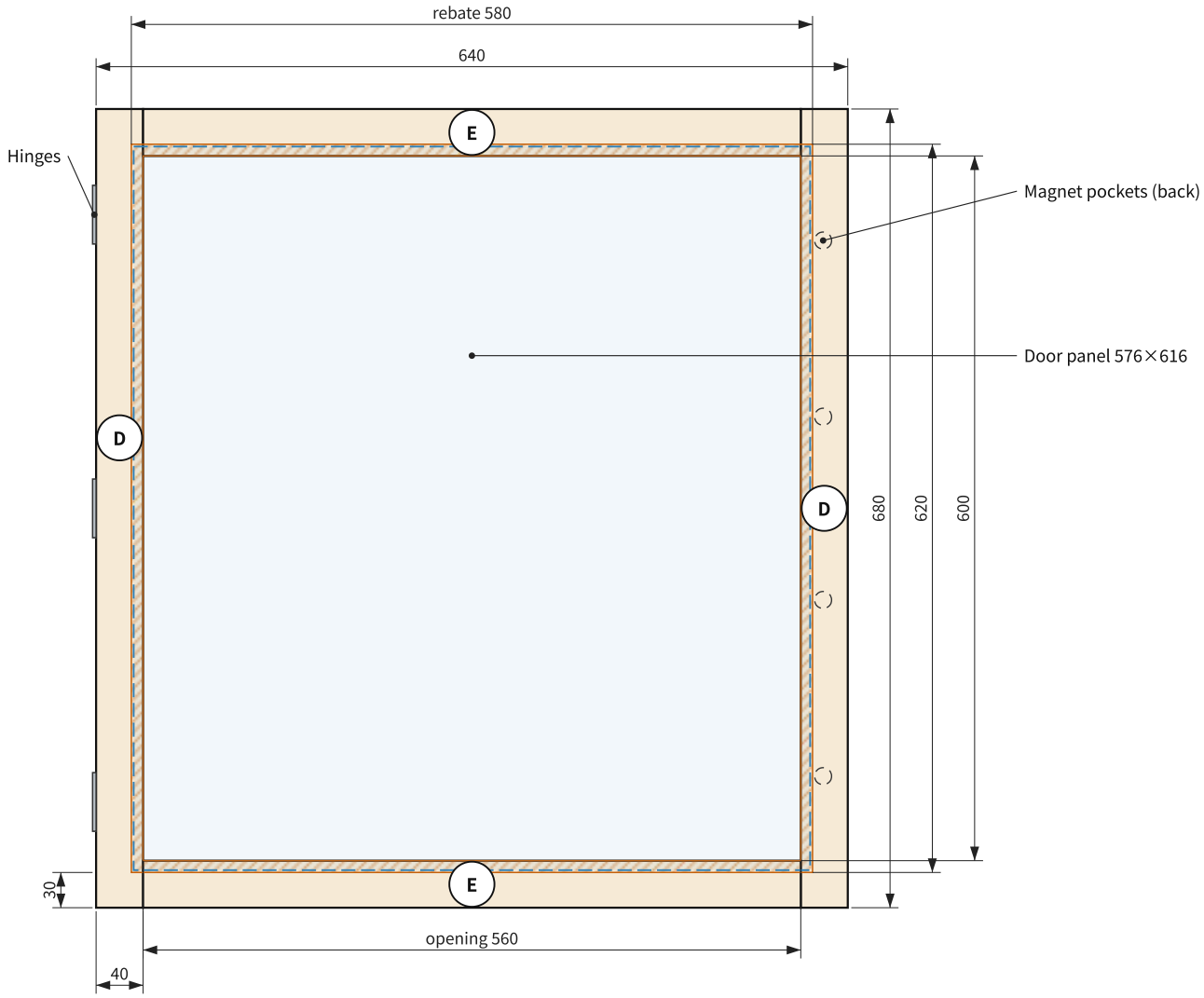
40 face outward, rebate on the bottom edge; 30 face up, rebate on the inner edge; full length.

Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

**Frame parts**

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 4 / 8 | 1:5 | mm | 2026-11-08 | v2 |

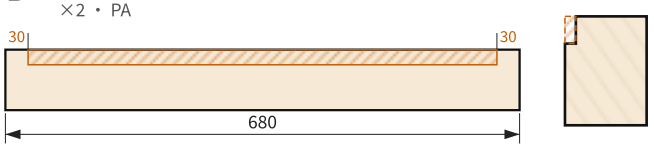
## Door frame, front view (from outside, scale 1:6)




The rebate is one continuous ring, 580×620 outside / 560×600 inside. Stiles D are stopped (30–650); rails E run full length; they meet at the corners. Magnet pockets and hinges are DIY, not shop work.

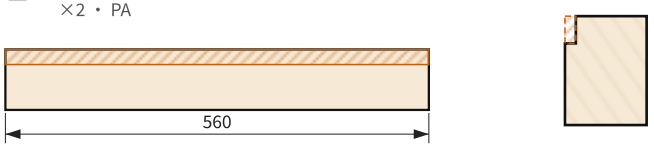
### Parts

**D** Door stile
×2 • PA




40 face to the front, rebate on the inner edge; stopped 30–650, 30 solid at each end.

**E** Door rail
×2 • PA



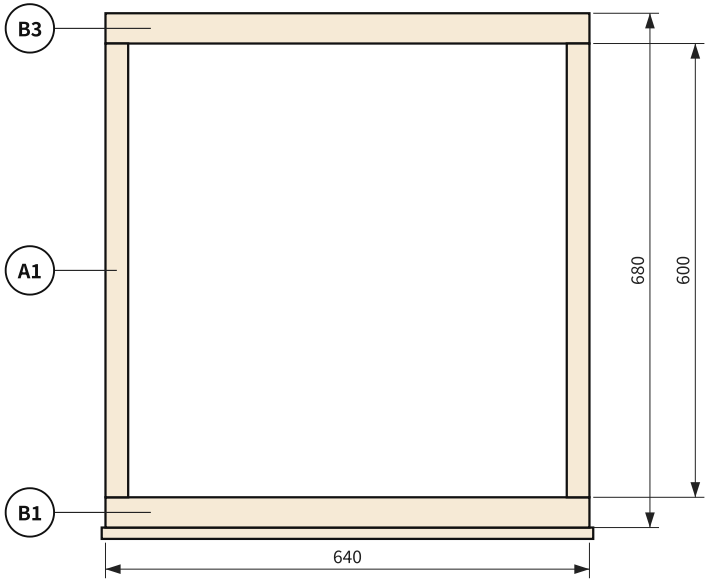

40 face to the front, rebate on the inner edge; full length.

Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

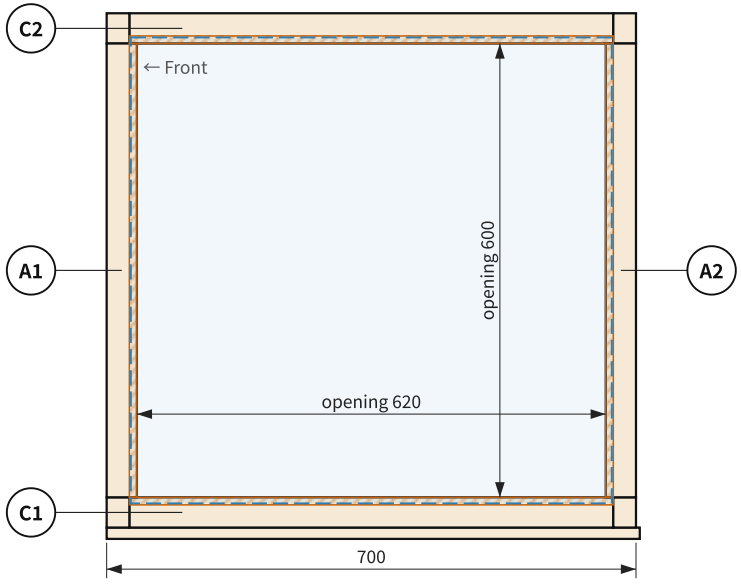
# Door frame

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 5 / 8 | 1:6 | mm | 2026-11-08 | v2 |

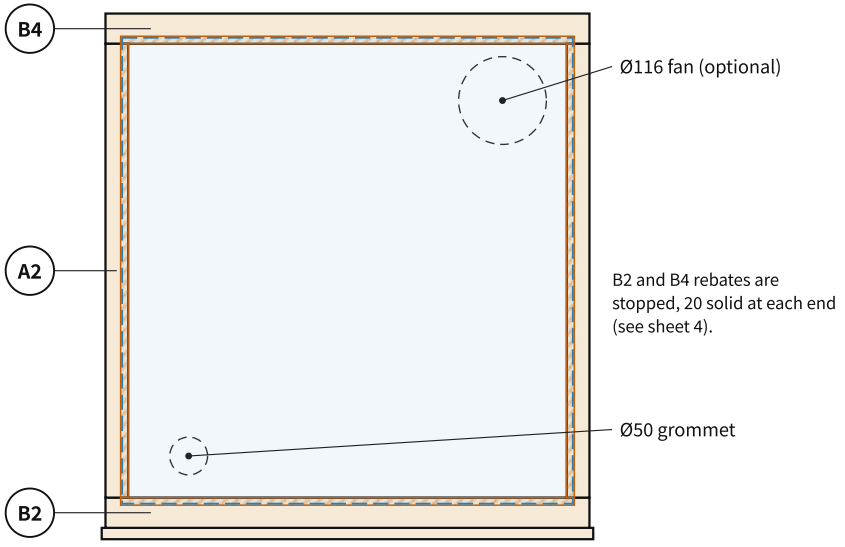
## Front view (door removed)



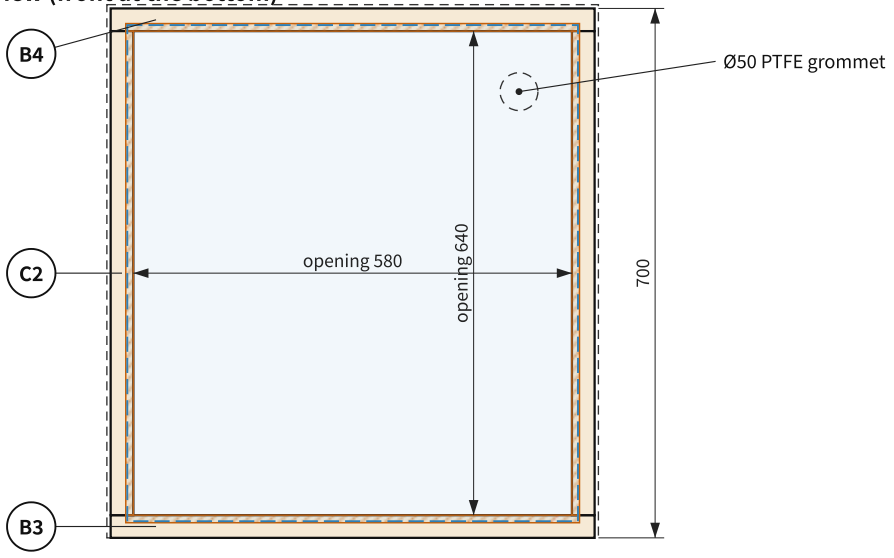

## Right side view (front on the left; left side mirrors it)




## Back view (from outside)




## Top view (front at the bottom)



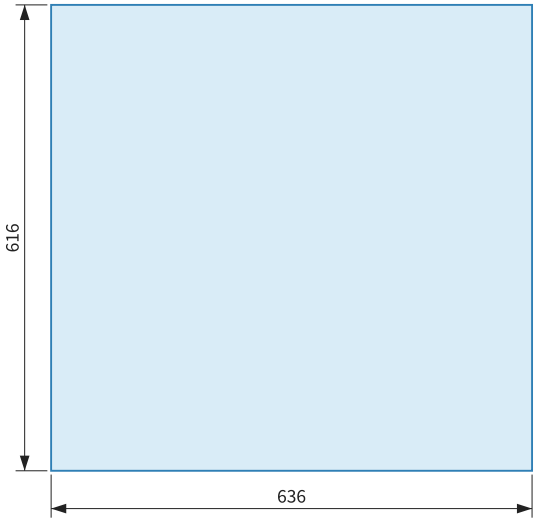

Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

# Frame assembly views

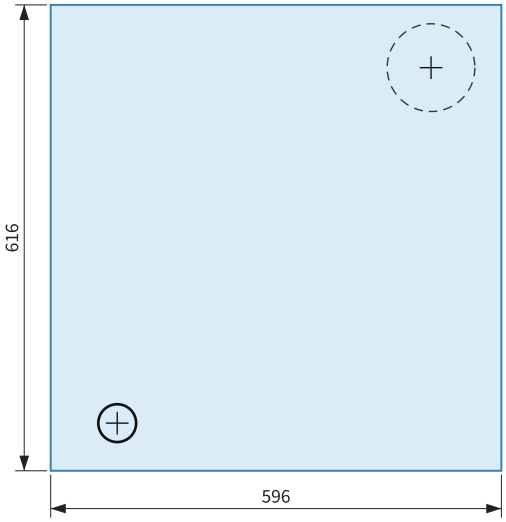
| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 6 / 8 | 1:10 | mm | 2026-11-08 | v2 |

## Clear panels (rebate version, scale 1:10, seen from outside)

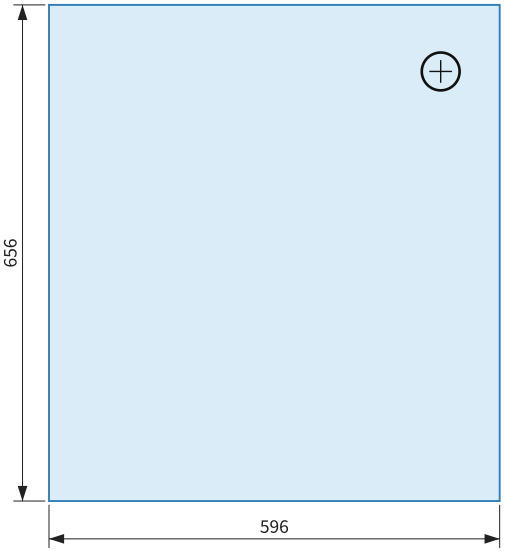
P1 Side panel (L/R) ×2



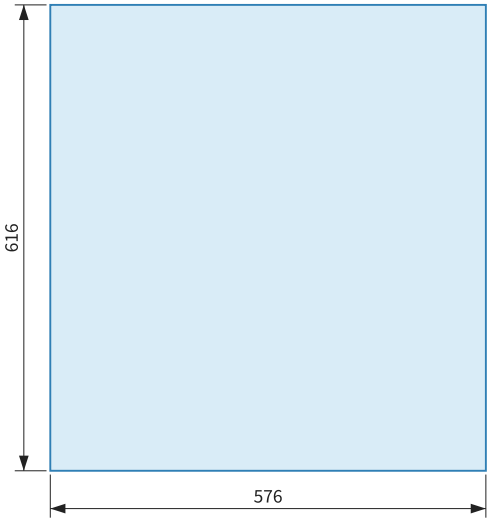

P2 Back panel ×1




P3 Top panel ×1




P4 Door panel ×1




| Panel | Hole | Position (to hole centre) |
|---|---|---|
| **P2** | Ø50 grommet hole | 88 from the left edge, 63 from the bottom (seen from outside; power-socket side) |
| **P2** | Ø116 fan hole (optional) | 93 from the right edge, 83 from the top |
| **P3** | Ø50 grommet hole | 78 from the right edge, 88 from the top (rear) edge; top of the drawing = back of the box |

- Material: 3 mm cast acrylic or solid polycarbonate (solid PC recommended for the door). Not twin-wall PC: it is not see-through.
- Size tolerance ±0.5, polished edges, protective film left on. No screw holes around the edges.
- Hole positions can be adjusted to suit the A1 power socket and the AMS lite. Drill the door-handle holes once you have the handle.

Bambu Lab A1 enclosure · external feed · rebate + bead (router needed)

**Clear panels**

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 7 / 8 | 1:10 | mm | 2026-11-08 | v2 |

## Appendix: screw-on version (no rebates)

Use this version if you have no router and the timber shop cannot cut rebates: no strip is rebated, and the panels lie on the outside of the frame, fixed with ST4×20 screws and nylon washers. Only this sheet differs; the other sheets still apply (ignore the rebates).

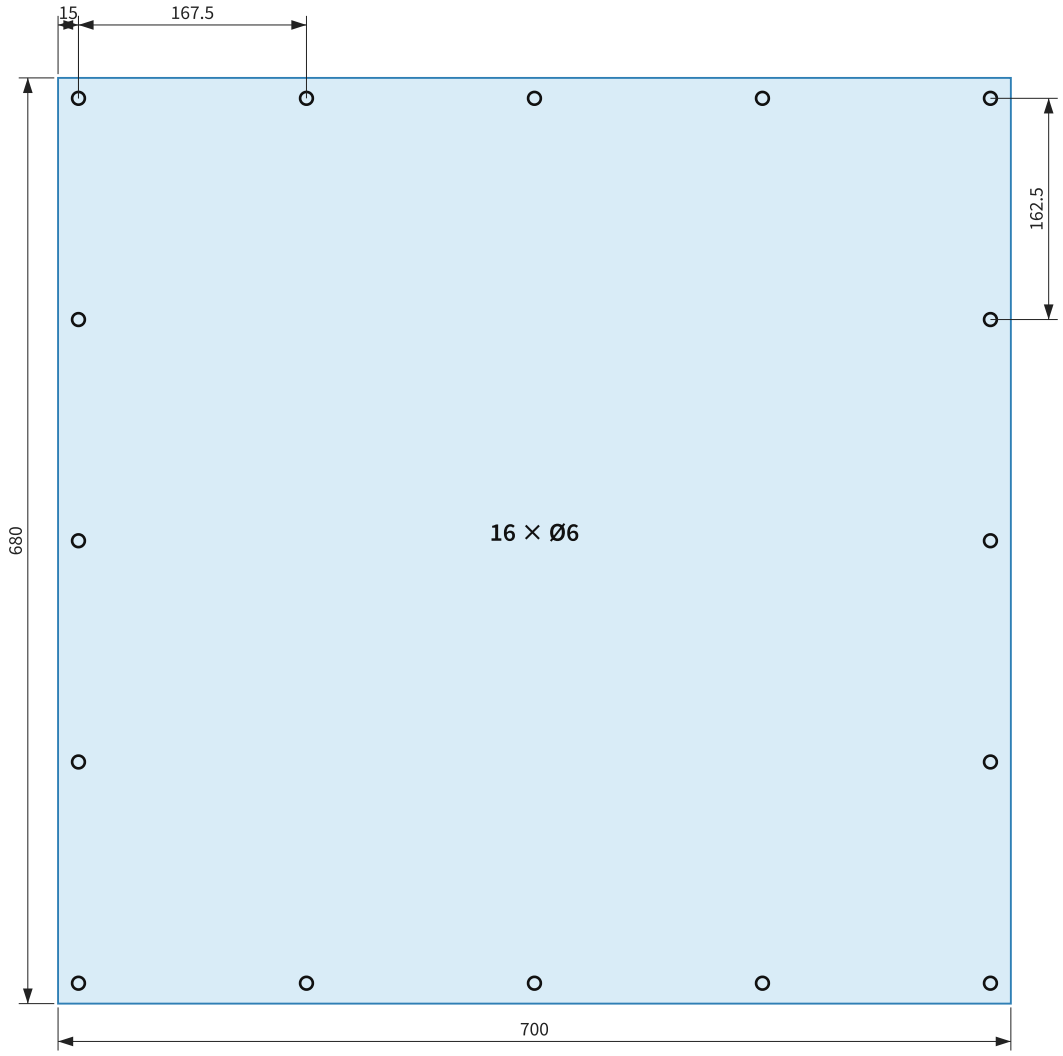
| ID | Part | Size | Qty |
|---|---|---|---|
| **A** | Post | 30×40×600 | 4 |
| **B** | Front/back rail | 30×40×640 | 4 |
| **C** | Side rail | 30×40×640 | 4 |
| **D** | Door stile | 30×40×683 | 2 |
| **E** | Door rail | 30×40×566 | 2 |
| **F** | Base board | 650×705×15 | 1 |

**Strip total 10 018 mm (6 × 2 m); no beads.**

| Panel | Size | Qty | Pitch (long / short) |
|---|---|---|---|
| **P1** | 700×680 | 2 | 167.5 / 162.5 |
| **P2** | 646×680 | 1 | 162.5 / 154 |
| **P3** | 646×703 | 1 | 168.25 / 154 |
| **P4** | 646×683 | 1 | 163.25 / 154 |

Each panel has 5 holes per edge (corners shared, 16 in all), Ø6, centres 15 from the edge. P2 Ø50 hole: 113 from the left and 95 from the bottom, seen from outside; optional Ø116 fan hole 118 from the right and 115 from the top. P3 Ø50 hole: 103 from the right, 113 from the top (rear) edge.

**P1 hole layout 1:5.5 (others alike)**




Bambu Lab A1 enclosure • external feed • rebate + bead (router needed)

**Appendix: screw-on version**

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 8 / 8 | — | mm | 2026-11-08 | v2 |